Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "prototype" and "constraint" to explain your model or result. Sentence starters are optional.

prototype: a version built to test a design

constraint: a requirement or limit that the design must meet

evidence: observations or measurements that support an idea

01 Bring back an earlier idea

A single-loop toy calculation gives 3 V induced emf. **Explain** why this alone cannot predict the effects on a whole power grid; name two missing features.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Sketch your proposed car. **Label** the battery, switch, motor, driven axle and wheels. Add a theme that will not obstruct moving parts.

Sentence starter: The … in my model represents … . Its label shows … .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

A supplied design has a 3 V supply and expected .40 A motor current.
Estimate electrical power.
Identify at least three reasons this is not all useful forward-motion power.

Word help: estimate = find an approximate value

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

05 Analyze the reasoning

Write one measurable design goal and one constraint.
Explain how you will measure the goal instead of judging only by appearance.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Choose a method

A circuit has 4 identical series resistors. Add two more at unchanged ideal supply voltage.
Compare new current with the original.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Create a fair first-test plan naming distance, timing method, number of trials and variables held fixed.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Choose a speed target and a reliability target for a 3 m course.
Design repeated trials and a rule for judging success without selecting only the fastest run.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3